EE2004 Digital Signal Processing

Tutorial 5 Answer Key

1. (a) $0.5^n\,u[n]$
   (b) $-0.25^n\,u[n] + 2\cdot 0.5^n\,u[n] - 2\,(n+1)\,0.5^n\,u[n] + (n+1)(n+2)\,0.5^n\,u[n]$

2. (a) $x[n] = \{\overset{\downarrow}{1},\ 4,\ 7,\ 10,\ 13,\ 16,\ 19,\ 22,\ 25,\ 28, \ldots\}$
   (b) $x[n] = \{\ldots, 11,\ 8,\ 5,\ 2,\ \overset{\downarrow}{0},\ 0,\ 0,\ \ldots\}$

3. (i) $\{\ldots, 0, 0, \overset{\downarrow}{0}, a - a^{-1}, a^2 - a^{-2}, a^3 - a^{-3}, \ldots\}$, i.e., $y[n] = (a^n - a^{-n})\,u[n-1]$
   (ii) $\{\ldots, a^3 - a^{-3}, a^2 - a^{-2}, a - a^{-1}, \overset{\downarrow}{0}, 0, 0, 0, \ldots\}$, i.e., $y[n] = (a^{-n} - a^n)\,u[-n]$

4. (a) $H(z) = \dfrac{1 - \frac{1}{2}\,z^{-1}}{1 + \frac{1}{2}\,z^{-1}}$ RoC: $|z| > \dfrac{1}{2}$
   (b) $0.5 < |z| < 2$
   (c) $\dfrac{3}{5}\ 2^n\,u[-n-1] + \dfrac{3}{5}\left(-\dfrac{1}{2}\right)^n\,u[n]$

5. Causal and stable

6. $-a^n\,u[-n-1]$

7. (a) $x[n] = \dfrac{1}{2^N}\,y[n+N]$ where $y[n] = \left\{\overset{\downarrow}{1}, 0, \binom{N}{1}, 0, \cdots, \binom{N}{N}\right\}$
   (b) $x[n] = \dfrac{1}{2^N}\,y[n+N/2]$ where $y[n] = \left\{\overset{\downarrow}{1}, \binom{N}{1}, \cdots, \binom{N}{N}\right\}$

8. (a) $h[n] = \begin{cases} \dfrac{(-1)^n}{n} & n \neq 0 \\ 0 & n = 0 \end{cases}$
   (b) $\pi^2/3$

9. (a) $\{4, 2, 1, 3, -4, 0, \overset{\downarrow}{-1}, 3, 2\}$
   (b) $\{2, -3, \overset{\downarrow}{-1}, 0, -4, -3, 1, -2, 4\}$

10. (a) conjugate even, (b) real and even, (c) real-valued

11. (a) purely imaginary with no symmetry
    (b) real-valued and odd
    (c) real-valued with no symmetry

12. $\{-1, 1, \overset{\downarrow}{1}\}$

13. 2